A complete C222 Diophantine equation

446 natural unknowns; degree 2,097,684.

Unknowns and front. Every free coordinate is in $\mathbb{N} = \{0, 1, 2, \ldots\}$:

$$h, e, g, \beta, F, x, N, \ell, z_0, z_t, z_B, \quad (\xi_i)_{i=0}^{8},$$
$$(D_i)_{i=0}^{3}, \quad (P_j)_{j=0}^{10}, \quad (r_j, s_j, w_j)_{j=0}^{10}, \quad (\theta_j)_{j=0}^{17}.$$

Each θ_j is the 21-tuple below. Every other name is a polynomial abbreviation; all bound indices are local to their displayed fixed ranges.

$$t = 16 + h, \quad y = F + 1 + e,$$
$$q = 2 + g, \quad B = 1 + \beta,$$
$$(C, a, b, c, d, S, T, u, v) = (1 + \xi_i)_{i=0}^{8},$$
$$H = 2 + C + a + b + c + d + S + T + u + v.$$

The whole numerical calculus. Define

$$I(a,b) = 2(a+b)(a+b+1) + 4b,$$
$$M(a,b) = I(a,b) + 1, \qquad A(a,b) = I(a,b) + 2,$$
$$E(a,b) = I(a,b) + 3, \qquad \nu(a) = I(a,1),$$
$$X(a,b) = \nu(A(a,\nu(b))), \qquad \Theta_a(z) = \nu(A(a,z)).$$

The first eight code polynomials are

$$\begin{aligned}
c_1 &= X(a, E(a,b)),\\
c_2 &= I(A(a, I(b,c)), I(A(a,b), A(a,c))),\\
c_3 &= I(E(a,b), I(M(c,a), M(c,b))),\\
c_4 &= I(A(a, A(b,c)), A(b, A(a,c))),\\
c_5 &= I(E(c,a), I(M(c,b), M(a,b))),\\
c_6 &= I(X(a, A(a,b)), b),\\
c_7 &= I(E(c,a), I(E(c,b), A(M(a,b), E(a,b)))),\\
c_8 &= I(I(I(c,d),a), I(I(a,c), I(b,c))).
\end{aligned}$$

The entire ninth code is

$$\begin{aligned}
R(b,k,z) &= A(b, I(z, \Theta_b(I(z, \nu(M(b,k)))))),\\
Q(a,b,k) &= A(k, I(M(k,a), M(k,b))),\\
C_0(a,b,k,z) &= I(Q(a,b,k), R(b,k,z)),\\
C_*(a,b,k,z) &= A\big(b, I(M(b,k),\\
&\qquad \Theta_a(I(z, \Theta_a(C_0(a,b,k,z)))))\big),\\
V(a,b,c) &= I(M(a,b), A(c, M(a,b))),\\
K(a,b,c) &= A(c, I(C_*(a,b,c,c), \nu(R(b,c,c)))),\\
c_9 &= \nu(I(V(a,b,c), K(a,b,c))).
\end{aligned}$$

Evaluate every c_i at (a,b,c,d), ignoring unused arguments. Put

$$\begin{aligned}
W_z &= 1 + Fz,\\
J &= \sum_{i=1}^{9} c_i + a + b + I(a,b) + A(a,b) + 1,\\
G &= ((c-a)^2(c-b)^2 - (d+1)^2)^2,\\
\mathcal{H} &= (T-S)^2 \prod_{i=1}^{8} (TW_{c_i} - S)^2\\
&\quad \cdot \big((TW_{c_9} - S)^2 + G\big)\\
&\quad \cdot (TW_b - SW_a W_{I(a,b)})^2\\
&\quad \cdot (TW_{A(a,b)} - SW_b)^2,\\
L_i &= \mathcal{H} + (x - S - (1 + (i+1)y)u)^2\\
&\qquad + (x - T - (1 + (i+2)y)v)^2.
\end{aligned}$$

Only L_C is used. The guarded branch retains both disequalities $c \neq a, c \neq b$ through its natural witness d and the printed G.

The 21-coordinate power polynomial. Every letter here is local to its copy, independent of the front and all other copies:

$$\theta = (Y, Z, X, U, S, R, a_0, y_0, f, d,$$
$$b_+, b_-, s_+, s_-, r_+, r_-, x_+, x_-, g, w_n, w_k).$$

For arguments (n, k, m) define only

$$A = a_0 + 2, \quad V = Y^2(d+1),$$
$$B = 1 + 4Y(f+1), \quad \tau = m + g + 1,$$
$$W = n + w_n,$$
$$\mathcal{N}(a,x,y) = x^2 + y^2 - 1 - a^2y^2.$$

In particular Y is free. Let $(R_1, \ldots, R_{11})$ be exactly

$$\big(Y - k - y_0,$$
$$\mathcal{N}(A,X,Y),\ \mathcal{N}(A,U,V),\ \mathcal{N}(B,S,R),$$
$$B + Ub_+ - A - Ub_-,\ S + Us_+ - X - Us_-,$$
$$R + 4Yr_+ - k - 4Yr_-,$$
$$X + Yn + \tau x_+ - YA - m - \tau x_-,$$
$$2An - \tau - n^2 - 1,\ W - k - w_k,\ \mathcal{N}(W+1, A, WZ)\big).$$

Set $\mathcal{P}(n,k,m;\theta) = \sum_{i=1}^{11} R_i^2$. All private coordinates, including both signs of every quotient, are natural.

Every argument slot. The full ordered table is

$$\begin{aligned}
(k_i)_{i=0}^{3} &= (t, t+1, y+1, x),\\
(U_j)_{j=0}^{10} &= (F^2 + t, C, a, b, c, d, S, T, u, v, J),\\
(\sigma_j)_{j=0}^{10} &= (0,0,1,1,1,1,1,2,2,3,3,1),\\
\mathcal{T} &= \big((2(t+1), t, F), (B, x, N), (BH, x, q),\\
&\qquad (q, k_i, D_i)_{i=0}^{3},\ (q+1, U_j, P_j)_{j=0}^{10}\big).
\end{aligned}$$

For $0 \le j \le 17$, write $(n_j, k_j', m_j) = \mathcal{T}_j$ in this order and pair it with θ_j. No variable exponent occurs.

The single equation.

$$\begin{aligned}
\mathcal{Q} = &\sum_{j=0}^{17} \mathcal{P}(n_j, k_j', m_j; \theta_j)\\
&+ (x - F - 1 - (y+1)z_0)^2\\
&+ (x - 1 - (1 + (t+1)y)z_t)^2\\
&+ (Bz_B - C - F^2 - 1)^2 + (L_C - N\ell)^2\\
&+ (P_0 - D_0(qw_0 + B) - r_0)^2\\
&+ \sum_{j=1}^{10} (P_j - D_{\sigma_j} N w_j - r_j)^2\\
&+ \sum_{j=0}^{10} (D_{\sigma_j} - r_j - s_j - 1)^2 = 0.
\end{aligned}$$

This prints the entire integer-coefficient polynomial: 224 squared residuals on 446 natural unknowns, of exact degree 2,097,684. For 446 positive unknowns, replace every free z independently by $p_z - 1$, $p_z \ge 1$. Relative to the retained ordinary Pell and C222/ZF/ZFC calibrations, absence of zeros is equivalent to syntactic Con(ZFC).